AI-PYTHON PROJECT
(Full stack)

INTELLIGENT TENDER PREPARATION SYSTEM
(IntelTPS)

Duration: 4 Weeks

Start Date: 16 October 2026

End Date 16 November 2026)

AI-ASSISTED TENDER PREPARATION PLATFORM

Software Requirements Specification, System Design and Implementation Plan

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1. PROJECT BRIEFING AND BACKGROUND

1.1 BRIEFING

AI Construction Project & Tender Assistant is a proposed cloud-based application designed to help South African construction businesses analyze tender documents, prepare estimates, organize compliance requirements and manage tender-related administrative activities.

The initial target market comprises small and medium-sized construction contractors, particularly electrical and plumbing subcontractors that regularly prepare quotations and tender submissions.

The application will use artificial intelligence (AI), document processing and conventional software automation to convert unstructured construction documents into structured, editable information.

The proposed system will enable users to upload tender documents, extract key information, identify submission requirements, process Bills of Quantities (BoQs), compare supplier quotations and produce draft tender preparation reports.

The system will not replace the contractor's estimator, quantity surveyor, accountant, legal adviser or procurement specialist. All critical pricing, compliance and submission decisions will remain under human control.

The product will follow a Software-as-a-Service business model in which customers pay a recurring subscription for access to the platform.

1.2 Background

Construction contractors frequently work with multiple document types, including:

- Tender invitations and instructions.
- Technical specifications and drawings.
- Bills of Quantities and pricing schedules.
- Supplier quotations.
- Company registration and contractor documentation.
- Tax, insurance and other supporting certificates.
- Project budgets, purchase orders and variation records.

These documents may arrive as scanned PDFs, spreadsheets, Word documents, email attachments or photographs.

When information must be manually extracted from these files, businesses may experience delays, inconsistent records, duplicated work, overlooked requirements and pricing mistakes.

Smaller contractors may not have dedicated estimating departments or sophisticated tender-management systems. Their processes may rely on spreadsheets, email, messaging applications and shared folders.

The proposed application aims to make selected tender-preparation activities more efficient without requiring businesses to replace their existing software.

1.3 Problem Statement

The project addresses five principal problems:

1. **Manual document analysis:** Tender packs can contain extensive instructions and supporting documentation that require careful review.
2. **BoQ processing difficulties:** Extracting quantities, units and item descriptions from inconsistent document formats can be time-consuming.
3. **Supplier quotation management:** Comparing supplier descriptions, specifications, prices and units can require substantial manual effort.
4. **Compliance tracking:** Contractors need to identify mandatory documents, deadlines and submission requirements.
5. **Fragmented information:** Tender documents, estimates, supplier quotations and task assignments may be stored in separate locations.

The project seeks to reduce these inefficiencies while preserving transparency and human accountability.

1.4 Project Vision

To provide an affordable, accessible and trustworthy AI-assisted tender preparation platform tailored to the workflows of South African construction businesses.

1.5 Project Objectives

The objectives are to:

- Reduce the time required to review tender documentation.
- Extract BoQ information into structured and editable tables.
- Assist with supplier quotation comparisons.
- Generate source-referenced tender requirement checklists.

- Identify missing information and potentially inconsistent values.
- Provide downloadable Excel and PDF reports.
- Establish a commercially viable recurring-revenue SaaS.
- Validate customer demand before investing in advanced integrations.

1.6 Project Scope

The initial Minimum Viable Product (MVP) will focus on tender preparation and estimating assistance.

Post-award project management, full accounting, live tender aggregation, automatic bid submission, construction scheduling and advanced forecasting are outside the initial scope.

2. BUSINESS CONTEXT, USERS AND REQUIREMENTS

2.1 Target Market

The primary customers will be South African construction businesses that regularly prepare quotations or tender submissions.

Initial customer segments include:

- Electrical subcontractors.
- Plumbing subcontractors.
- Painting contractors.
- Small building contractors.
- Specialist construction suppliers.
- Independent estimators and quantity surveying practices.

The initial launch should focus on one trade to reduce development complexity and make customer validation more effective.

Electrical subcontractors are proposed as the initial niche, subject to market validation.

2.2 User Roles

A. Business Owner or Administrator

The administrator manages the business account, users, subscription, company information and permissions.

Responsibilities include:

- Creating and managing workspaces.
- Inviting team members.
- Maintaining approved company information.
- Configuring pricing and markup defaults.
- Reviewing activity logs.
- Managing document access.

B. Estimator or Tender Officer

The estimator is the primary application user.

Responsibilities include:

- Uploading tender documents.
- Reviewing AI-extracted requirements.
- Correcting BoQ extraction errors.
- Importing supplier quotations.
- Comparing prices and specifications.
- Preparing draft estimates and reports.
- Tracking outstanding tender requirements.

C. Project Manager

The project manager monitors tender progress and coordinates work.

Responsibilities include:

- Viewing tender deadlines.
- Assigning tasks.
- Monitoring outstanding requirements.
- Reviewing approved estimates.
- Accessing tender preparation reports.

D. Read-Only Reviewer

A reviewer may inspect documents, estimates and checklists without modifying approved information.

This role is useful for managers, directors or external professional advisers.

E. Platform Administrator

The platform administrator manages application health, subscription operations, support and system configuration.

Access to customer documents must be restricted, logged and limited to legitimate operational needs.

2.3 User Stories

The following user stories define the main expected outcomes.

US-01: Upload a tender pack

As an estimator, I want to upload a tender pack so that the system can extract useful information.

Acceptance criteria:

- The system accepts supported file types.
- Unsupported files produce a clear error.
- Each upload receives a processing status.
- The original file remains linked to its extracted data.

US-02: Extract a BoQ

As an estimator, I want BoQ lines extracted into a table so that I can review quantities and prices efficiently.

Acceptance criteria:

- Each extracted line has a description, unit and quantity where available.
- Source page or worksheet information is retained.
- Uncertain fields are flagged.
- Users can correct extracted values.

US-03: Review tender requirements

As a tender officer, I want a checklist of mandatory requirements so that I can organise the submission process.

Acceptance criteria:

- Requirements can be edited and assigned.
- Source references are available.
- Each requirement has a status.
- The checklist distinguishes extracted information from verified information.

US-04: Compare supplier quotations

As an estimator, I want to compare supplier quotations against BoQ items so that I can identify suitable pricing options.

Acceptance criteria:

- Supplier and quotation details are recorded.
- Prices and units are visible.
- Potential matches include supporting evidence.
- Ambiguous matches require human confirmation.

US-05: Export estimate

As a contractor, I want to export an approved draft estimate so that I can continue using my existing spreadsheet workflow.

Acceptance criteria:

- The exports include approved line items.
- Calculations are performed by deterministic code.
- The document identifies its version and generation date.
- Draft outputs are clearly labelled.

2.4 Stakeholders

The main stakeholders are the developer, contractor owners, estimators, project managers, subcontractors, suppliers and relevant professional advisers.

Their needs may differ. Contractors prioritize speed and cost, while estimators prioritize numerical accuracy, traceability and control.

The system must balance usability with reliability.

3. FUNCTIONAL REQUIREMENTS

3.1 Authentication and User Management

FR-01: The system shall allow authorized users to register or join a business workspace.

FR-02: The system shall authenticate users before granting access to protected resources.

FR-03: The system shall support role-based access control.

FR-04: The system shall allow administrators to invite, deactivate and manage users.

FR-05: The system shall enforce tenant isolation so that one business cannot access another business's data.

3.2 Tender Workspace Management

FR-06: Users shall be able to create a tender workspace containing a tender title, reference number, client, location, closing date and status.

FR-07: Users shall be able to attach multiple documents to a tender workspace.

FR-08: Users shall be able to update tender details and track the current processing stage.

FR-09: The system preserves document versions and identify which version was used for analysis.

FR-10: Users shall be able to archive complete or abandoned tenders.

3.3 Document Upload and Processing

FR-11: The system shall support PDF, XLSX, CSV and DOCX files where applicable.

FR-12: The system shall validate file type, size and upload permissions.

FR-13: The system shall extract embedded text from supported documents.

FR-14: The system shall support OCR for scanned documents.

FR-15: The system shall record processing status, errors and timestamps.

FR-16: The system shall allow users to retry failed processing jobs.

FR-17: The system shall treat uploaded document content as untrusted input.

3.4 Tender Intelligence

FR-18: The system shall extract potential tender deadlines, compulsory meetings, submission instructions and eligibility requirements.

FR-19: Each extracted requirement shall retain a reference to the source document and page or worksheet where available.

FR-20: Users shall be able to edit, approve or reject extracted requirements.

FR-21: The system shall allow users to classify requirements as mandatory, conditional or informational.

FR-22: The system shall generate a checklist of requirements requiring further action.

FR-23: The system shall not represent an AI-generated checklist as an authoritative compliance determination.

3.5 BoQ Extraction and Estimating

FR-24: The system shall extract BoQ line numbers, descriptions, units, quantities and existing prices where available.

FR-25: Users shall be able to edit extracted line items.

FR-26: The system shall preserve original extracted values separately from user corrections.

FR-27: The system shall support user-defined material, labor and markup fields where appropriate.

FR-28: All financial calculations shall use deterministic application logic.

FR-29: The system shall flag missing quantities, units, prices and unusual values.

FR-30: Users shall be able to produce a draft estimate without automatically submitting it to a procurement authority.

3.6 Supplier Quotation Comparison

FR-31: Users shall be able to create supplier records.

FR-32: Users shall be able to upload supplier quotations.

FR-33: The system shall extract available supplier names, item descriptions, quantities, units, prices and quotation dates.

FR-34: The system shall suggest possible matches between supplier items and BoQ lines.

FR-35: Users shall confirm or reject suggested matches.

FR-36: The system shall display differences in units, specifications, price and quotation validity where available.

FR-37: The system shall not automatically select a supplier solely because its displayed price is lowest.

3.7 Reports and Export

FR-38: The system shall generate Excel exports of approved BoQ data.

FR-39: The system shall generate a tender checklist report.

FR-40: The system shall include relevant source references and generation dates in reports.

FR-41: The system shall distinguish draft, reviewed and approved outputs.

FR-42: The system shall record export events in the audit trail.

3.8 Notifications and Task Management

FR-43: Users shall be able to assign tender requirements to team members.

FR-44: The system shall display approaching deadlines.

FR-45: The system shall support in-app reminders and, when configured, email notifications.

FR-46: Users shall be able to update task statuses.

3.9 Subscription and Usage Controls

FR-47: The system shall record subscription status and plan limits.

FR-48: The system shall enforce configurable document-processing limits.

FR-49: The system shall prevent unauthorized access to paid features.

FR-50: The system should provide clear usage information to the customer.

4. NON-FUNCTIONAL REQUIREMENTS AND BUSINESS RULES

4.1 Performance

The MVP should provide responsive interactions for normal administrative tasks.

Proposed targets:

- File upload confirmation: within reasonable time (five seconds) for typical files on a suitable connection.
- Document extraction: asynchronous processing with visible progress and an estimated completion state.
- Excel export: completed within appropriate time (ten seconds) for ordinary tender workspaces, excluding unusually large datasets.

These targets must be tested and adjusted based on hosting capacity and actual customer usage.

4.2 Reliability

The application shall:

- Recover gracefully from document-processing failures.
- Avoid creating duplicate records when a job is retired.

- Record errors without exposing secrets or sensitive document contents in logs.
- Provide database backup and restoration procedures.
- Keep processing status visible to users.

4.3 Security

The application shall implement:

- HTTPS for production traffic.
- Secure password handling or a trusted authentication provider.
- Role-based authorization.
- Tenant-level access controls (if possible).
- Restricted file downloads.
- Environment-based secret management.
- Input validation and output encoding.
- Rate limiting for sensitive endpoints.
- Audit logging for important changes.
- Dependency and vulnerability monitoring.

Authorization must be enforced on the backend. Hiding an interface element is not sufficient security.

4.4 Privacy and Data Protection

The product may process personal information, confidential bids, supplier pricing and commercially sensitive company records.

The developer shall assess applicable obligations under South Africa's Protection of Personal Information Act (POPIA).

The product should provide:

- A privacy notice.
- A documented purpose for data collection.
- Retention and deletion procedures.
- Access controls for customer information.
- Written agreements with relevant service providers where required.
- A process for handling data access and correction requests.

- An incident-response procedure.
- A documented assessment of cross-border data transfers where relevant.

The application should not send confidential tender information to an AI provider without appropriate contractual, privacy and security assessment.

Reference: South African Government — Protection of Personal Information Act.

4.5 Usability

The interface shall be used by people with ordinary office-computing skills.

The system should provide:

- Clear labels and validation messages.
- Mobile-responsive pages.
- Editable extracted values.
- Filters and search.
- Clear draft, processing, failed and approved statuses.
- Downloadable reports.
- Keyboard-accessible controls.

4.6 Maintainability

The backend shall separate business logic, data access, document processing and AI integration.

The project shall use automated tests, database migrations, environment-specific configuration and documented deployment procedures.

4.7 Scalability

The MVP shall use a modular monolith and a separate background worker where appropriate.

Architecture should permit future migration to more specialized services without requiring immediate microservices architecture.

4.8 Core Business Rules

BR-01: Extracted information is unverified until reviewed or approved by an authorized user.

BR-02: Every important AI-extracted requirement should retain source provenance wherever technically possible.

BR-03: Financial calculations must be performed using deterministic code and appropriate decimal precision.

BR-04: Ambiguous supplier matches must not be silently accepted.

BR-05: Each business workspace may access only its own authorized data.

BR-06: Tender deadlines and requirements must be checked against the current authoritative tender documents.

BR-07: The application shall not guarantee tender compliance, tender success or the accuracy of every AI-generated result.

BR-08: The system shall not submit bids automatically in the MVP.

5. SYSTEM ARCHITECTURE AND TECHNOLOGY STACK

5.1 Architectural Overview

The system will follow modular monolithic architecture.

The frontend will communicate with a Python API. The API will manage authentication, business rules, permissions and database operations.

Document extraction and AI analysis will run asynchronously for longer tasks.

PostgreSQL will store structured application data. Object storage will hold original documents and generated files.

5.2 Proposed Technology Stack

Layer	Technology	Purpose
Frontend	Next.js and TypeScript	User interface
UI styling	Tailwind CSS	Responsive layouts
Backend	Python and FastAPI	REST API
Validation	Pydantic	Data validation
ORM Object Relation Mapping	SQLAlchemy Use for database interactions	Database access
Database	PostgreSQL	Structured records
Migrations	Alembic	Schema evolution
PDF extraction	PyMuPDF or pdfplumber	Text extraction
Spreadsheet processing	openpyxl and pandas	XLSX and CSV handling

OCR	Tesseract initially	Scanned-document text
AI integration	LLM API with structured outputs	Extraction and classification
Background jobs	Simple worker initially; Celery/Redis later	Asynchronous processing
File storage	Data lake gen2, Blob	Original and generated documents
Deployment	Docker	Consistent environments
Testing	pytest and frontend tests	Quality assurance

5.3 Main Application Modules

The backend should be divided into the following logical modules:

1. Authentication and user management.
2. Business workspaces and permissions.
3. Tender and project management.
4. Document storage and processing.
5. AI extraction and validation.
6. BoQ and estimating.
7. Supplier management and quotation comparison.
8. Checklist and task management.
9. Reporting and exports.
10. Subscription and usage management.
11. Audit logging and administration.

Each module should have clear interfaces and limited responsibility.

5.4 Request Processing Flow

A typical document-processing request follows this sequence:

1. The user uploads a document.
2. The API validates the request and checks permissions.
3. The file is stored securely.
4. A document record and processing job are created.

5. A worker extracts text or spreadsheet data.
6. The AI component produces structured candidate data.
7. The validation layer checks the output.
8. The extracted result is stored with source metadata.
9. The frontend displays the result for review.
10. The user approves or corrects the information.
11. Approved data becomes available for reporting and export.

The system must handle processing failure without losing the original file.

5.5 Deployment Environments

The project should have three environments:

Development: Local Docker environment with synthetic documents.

Staging: A production-like environment for testing deployments and integrations.

Production: Restricted customer access, managed database, backups, monitoring and secure file storage.

Development and production credentials must never be shared.

5.6 Why Not Microservices?

Microservices would introduce unnecessary deployment, monitoring and operational complexity for a low-budget MVP.

A modular monolith allows the developer to build and maintain the system more efficiently while keeping future separation possible.

6. DATABASE DESIGN AND API SPECIFICATION

6.1 Core Database Entities

Users

Stores identity and account information.

Suggested fields:

- id
- email
- name

- auth_provider_id
- created_at
- status

Organizations

Represents a contractor or business workspace.

Suggested fields:

- id
- name
- registration_number
- contact_email
- created_at

Organization Memberships

Associates users with organizations and roles.

Suggested fields:

- id
- organisation_id
- user_id
- role
- created_at

Tenders

Represents an individual tender opportunity.

Suggested fields:

- id
- organisation_id
- title
- reference_number
- client_name
- location

- closing_at
- status
- created_by
- created_at
- updated_at

Documents

Stores metadata for uploaded documents.

Suggested fields:

- id
- organisation_id
- tender_id
- filename
- storage_key
- mime_type
- file_size
- checksum
- processing_status
- uploaded_by
- created_at

Extraction Jobs

Tracks asynchronous processing.

Suggested fields:

- id
- document_id
- job_type
- status
- attempt_count
- error_code

- started_at
- completed_at

Tender Requirements

Stores requirements extracted from tender documents.

Suggested fields:

- id
- tender_id
- description
- requirement_type
- mandatory_status
- source_document_id
- source_page
- verification_status
- assigned_to
- due_at

BoQ Items

Stores structured line items.

Suggested fields:

- id
- tender_id
- item_number
- description
- unit
- quantity
- unit_price
- source_document_id
- source_page
- review_status

Suppliers

Suggested fields:

- id
- organisation_id
- name
- contact_email
- contact_phone

Supplier Quotations

Suggested fields:

- id
- organisation_id
- supplier_id
- document_id
- quotation_reference
- quotation_date
- valid_until
- currency
- status

Supplier Quotation Items

Suggested fields:

- id
- quotation_id
- description
- unit
- quantity
- unit_price
- source_reference

Tasks

Suggested fields:

- id
- tender_id
- title
- description
- assigned_to
- status
- due_at

Audit Events

Suggested fields:

- id
- organisation_id
- user_id
- entity_type
- entity_id
- action
- timestamp
- metadata

Audit metadata should avoid unnecessarily storing sensitive document contents.

6.2 Data Relationships

- An organization has multiple users through memberships.
- An organization has multiple tenders.
- A tender has multiple documents, requirements, BoQ items and tasks.
- A supplier has multiple quotations.
- A quotation has multiple quotation items.
- A document may have multiple extraction jobs.
- Important user actions create audit events.

All organization-owned records must be associated with an organization identifier.

6.3 API Endpoint Proposal

Method	Endpoint	Purpose
POST	/api/v1/tenders	Create tender
GET	/api/v1/tenders	List tenders
GET	/api/v1/tenders/{id}	Retrieve tender
PATCH	/api/v1/tenders/{id}	Update tender
POST	/api/v1/tenders/{id}/documents	Upload document
GET	/api/v1/documents/{id}	Retrieve document metadata
POST	/api/v1/documents/{id}/process	Request processing
GET	/api/v1/jobs/{id}	Check processing status
GET	/api/v1/tenders/{id}/requirements	List requirements
PATCH	/api/v1/requirements/{id}	Review requirement
GET	/api/v1/tenders/{id}/boq	Retrieve BoQ items
PATCH	/api/v1/boq-items/{id}	Update BoQ item
POST	/api/v1/quotations	Register quotation
GET	/api/v1/tenders/{id}/comparisons	Retrieve comparisons
POST	/api/v1/tenders/{id}/exports	Generate export

The API must validate request bodies, enforce permissions and return consistent error responses.

7. AI DOCUMENT-PROCESSING DESIGN

7.1 AI Objectives

AI will assist with:

- Extracting structured data from unstructured documents.
- Identifying tender requirements.
- Classifying requirements into useful categories.

- Matching supplier descriptions to BoQ items.
- Producing draft summaries and reports.
- Identifying information that requires human attention.

AI will not independently approve final prices, certify compliance or submit bids.

7.2 Document Processing Pipeline

Stage 1: File Validation

Check file type, file size, upload permissions and processing limits.

Files should be scanned or otherwise checked using suitable security controls before being processed.

Stage 2: Text and Table Extraction

For PDFs, attempt embedded-text extraction first.

For scanned documents, use OCR.

For spreadsheets, read cell values and worksheet structure directly.

Where tables are complex, preserve row order, column labels and source locations.

Stage 3: AI Extraction

Send the minimum necessary document content to the chosen AI provider.

Request structured output that conforms to a defined schema.

Example:

```
{
  "item_number": "A12",
  "description": "Supply and install electrical cable",
  "unit": "m",
  "quantity": "250",
  "source_page": 18,
  "needs_review": true
}
```

This is an illustrative extraction, not a real tender item.

The system must distinguish missing information from a value of zero. It must not invent quantities, units or prices.

Stage 4: Validation

Validate the extracted structure using Pydantic and business rules.

Examples:

- Quantity must be numeric where required.
- Unit must be present or explicitly marked missing.
- Source references must point to a known document.
- Prices must use an explicitly defined currency.
- Unexpected or contradictory values must be flagged.

Stage 5: Human Review

Display the extracted value alongside the original document or source reference.

Allow users to edit, approve or reject each item.

Record the original extracted value, corrected value, reviewer and timestamp.

Stage 6: Export

Only reviewed records should be included in exports designated as approved.

Draft exports must be clearly labelled.

7.3 Supplier Matching

Supplier item matching should use multiple signals:

- Text similarity.
- Product codes.
- Dimensions and technical specifications.
- Units of measurement.
- Manufacturer and model.
- Price and quotation date.

Text similarity alone is insufficient.

For example, two cables may have similar descriptions but different ratings or specifications. The system must flag such cases instead of assuming equivalence.

7.4 Financial Calculation Rules

Use Python's Decimal type or equivalent database numeric types for monetary calculations.

A basic extension formula is:

line_total = quantity × unit_price

Any VAT, labour, overhead or markup calculations must follow explicit, configurable business rules.

The application must not assume that every amount is VAT-inclusive or VAT-exclusive.

7.5 AI Evaluation

Create a manually verified test set containing representative tender packs, BoQs and supplier quotations.

Measure:

- Field-level extraction accuracy.
- Quantity and unit accuracy.
- Requirement extraction precision and recall.
- Supplier match accuracy.
- Percentage of fields requiring correction.
- Average processing cost per document.
- Average time saved per tender.

Critical fields such as quantities, units, deadlines and technical specifications should receive stricter acceptance criteria than general summaries.

7.6 Prompt Injection and Untrusted Documents

Uploaded documents may contain malicious or irrelevant instructions.

Document text must be treated as data, not as instructions that can override application rules.

The AI component must not receive unrestricted tools or permissions to execute arbitrary code, access unrelated customer records or submit tenders.

8. USER INTERFACE, SECURITY AND COMPLIANCE

8.1 Main Screens

Dashboard

Displays:

- Active tenders.

- Upcoming closing dates.
- Requirements awaiting review.
- Processing failures.
- Recently updated estimates.

Tender List

Allows users to search, filter, create and archive tenders.

Tender Workspace

Contains tender details, documents, extracted requirements, BoQ items, tasks and exports.

Document Review Screen

Displays the source document and extracted fields side by side.

BoQ Editor

Provides an editable table containing item numbers, descriptions, units, quantities, unit prices and calculated totals.

Supplier Comparison Screen

Displays supplier quotations, matching evidence, unit differences and price comparisons.

Checklist Screen

Displays requirements, source references, verification status, assigned users and deadlines.

Settings Screen

Allows authorized users to manage their organization, membership, subscription and preferences.

8.2 User Experience Principles

The product should prioritize clarity and accuracy over decorative AI features.

Every extracted value should have a visible review state.

Users should be able to identify:

- What the system extracted.
- Where the value came from.
- What has been verified.
- What remains uncertain.

- Who changed the information.

8.3 Proposed Security Controls

The MVP should implement:

1. Backend authorization for every protected operation.
2. Organization-level access checks.
3. Secure file storage with restricted access.
4. Short-lived signed download links where appropriate.
5. Encryption in transit.
6. Secure management of credentials and API keys.
7. Rate limits and file-upload restrictions.
8. Database backups and tested restoration.
9. Audit trails for important changes.
10. Dependency updates and vulnerability reviews.

File names and metadata must not be trusted as proof of a file's actual content type.

8.4 South African Procurement Context

The platform should support the analysis of construction procurement documents, including requirements derived from applicable tender conditions and procurement guidance.

The Construction Industry Development Board (cidb) publishes procurement prescripts and guidance relevant to public-sector construction procurement.

Official resources:

- eTenders: https://www.etenders.gov.za/Home/
- cidb procurement prescripts: https://www.cidb.org.za/clients/procurement-prescripts/cidb-prescripts/
- cidb tender evaluation guidance: https://www.cidb.org.za/clients/procurement-guides/evaluating-tenders/

Procurement requirements can change. The platform should record source documents, document versions and date of analysis.

It should not hard-code a single set of rules and represent them as universally applicable to all public and private tenders.

8.5 POPIA Considerations

The platform may process personal information relating to employees, supplier contacts and customer representatives.

The developer should assess applicable POPIA requirements and obtain professional advice where needed.

Controls should include appropriate security safeguards, access restrictions, retention policies, service-provider agreements and procedures for responding to security incidents.

Official reference: https://www.gov.za/documents/protection-personal-information-act

This document is a technical planning specification, not legal advice or certification of regulatory compliance.

9. TESTING, DEPLOYMENT AND BUDGET

9.1 Testing Strategy

Unit Testing

Test:

- Financial calculations.
- Data validation.
- Permission checks.
- Duplicate detection.
- BoQ calculations.
- Requirement classification rules.
- Export formatting.

Integration Testing

Test the complete flow from document upload through extraction, validation, review and export.

Test failures such as:

- Corrupted files.
- OCR errors.
- AI provider timeouts.
- Invalid structured output.

- Database interruptions.
- Duplicate processing requests.
- Expired download links.

Security Testing

Test that users cannot:

- Read another organisation's records.
- Access files by guessing identifiers.
- Modify records without permission.
- Bypass subscription limits.
- Execute arbitrary file content through uploads.

User Acceptance Testing

Recruit three to five contractors to test real workflows using documents they have permission to share.

Record usability problems, correction rates and time saved.

9.2 Proposed MVP Acceptance Criteria

The MVP may be considered ready for a controlled pilot when:

- Users can create accounts and manage tender workspaces.
- Supported files can be uploaded and processed.
- Extracted BoQ items can be reviewed and corrected.
- Source references are available for critical extracted fields.
- Financial totals are calculated deterministically.
- Supplier quotation comparisons can be reviewed.
- Checklist items can be assigned and tracked.
- Excel exports contain the approved information.
- Tenant isolation tests pass.
- Backups and restoration have been tested.
- Known limitations are documented for pilot customers.

AI accuracy targets should be defined using the manually verified test set before production use.

9.3 Indicative Starting Budget

The target is to keep the initial cash expenditure below R2,000 by developing the system personally.

Expense	**Initial estimate**
Development tools and open-source libraries	R0
Domain and basic hosting	R200–R600
AI and OCR testing	R150–R500
Storage, email and contingency	R150–R400
Target initial expenditure	**R500–R1,500**

These are planning estimates, not live supplier quotes. Actual costs depend on provider's pricing, document volume, file size and the availability of free tiers.

The budget excludes developer time, legal advice, paid procurement-data licenses and extensive production infrastructure.

9.4 Cost-Control Strategy

To remain within budget:

- Use local development and Docker.
- Begin with open-source parsing and OCR tools.
- Use a low-cost managed database where appropriate.
- Limit AI calls and document sizes.
- Process documents asynchronously.
- Cache safe, repeatable processing results when appropriate.
- Avoid building a live tender aggregation engine initially.
- Use customer-uploaded documents for the first version.
- Delay expensive third-party integrations until there is demonstrated demand.

9.5 Deployment Procedure

A basic production deployment should follow this sequence:

1. Run automated tests.
2. Build and verify the application container.

3. Apply reviewed database migrations.
4. Deploying to staging.
5. Test authentication, document processing and exports.
6. Back up production data before risky changes.
7. Deploy the approved release.
8. Monitor errors and processing failures.
9. Verify rollback and restoration procedures.

Deployment credentials must be stored securely and excluded from source control.

10. IMPLEMENTATION ROADMAP, RISKS AND SUCCESS METRICS

10.1 Implementation Roadmap

Day 1 - 4: Discovery and Requirements Validation

- Interview 5–10 contractors.
- Select one trade.
- Collect representative, permissioned sample documents.
- Map the existing estimating workflow.
- Finalise MVP acceptance criteria.

Deliverable: Validated workflow and initial product specification.

Day 5 – 12: Core Application and Extraction

- Create the repository and development environment.
- Implement authentication and organisation isolation.
- Build tender and document management.
- Implement PDF, spreadsheet and OCR processing.
- Create the first BoQ extraction pipeline.

Deliverable: A working extraction prototype with editable results.

Day 13 - 17: Estimating

- Build the BoQ editor.
- Add deterministic calculations.
- Implement data validation.
- Create supplier records and quotation imports.
- Add basic supplier matching.

Deliverable: A usable estimating workspace.

Day 18 - 21: Tender Intelligence

- Extract deadlines and submission requirements.
- Add source references.
- Implement requirement verification.
- Build the checklist and task assignment interface.

Deliverable: A tender preparation workspace.

Day 22 - 25: Reports and Production Readiness

- Implement Excel exports.
- Add audit history.
- Improve access controls.
- Configure backups, logging and usage limits.
- Conduct initial security testing.

Deliverable: A controlled pilot release.

Weeks 26–30: Pilot and Deployment Validation

- Onboard three to five contractors.
- Measure processing accuracy and time savings.
- Identify the most valuable features.
- Test initial pricing.

Deliverable: Evidence supporting the decision to continue, pivot or stop.

10.2 Risk Register

Risk	Impact	Mitigation
Incorrect AI extraction	High	Source references, validation and human review
Incorrect supplier matching	High	Specification checks and explicit approval
Document-format variability	Medium–high	Supported-format limits and robust testing
Data breach	High	Access controls, encryption, backups and incident response
Procurement rules change	Medium–high	Versioned sources and periodic review
Scope becomes too broad	High	Maintain a strict MVP backlog
Customer-specific customization	Medium–high	Standardize workflows before adding bespoke features

10.3 Key Performance Indicators

The business should measure both technical performance and customer value.

Product Metrics

- Number of documents processed.
- Extraction accuracy by field.
- Percentage of fields corrected by users.
- Average processing time.
- Failed processing rate.
- Cost per processed document.
- Number of active tender workspaces.

Value Metrics

- Time saved per BoQ.
- Time saved preparing a tender checklist.
- Number of missing requirements identified before submission.
- Reduction in manual data entry.
- Frequency of repeat use.

The system should not claim that it increases tender success rates unless that outcome is measured using appropriate evidence.

APPENDIX A — INITIAL DEVELOPMENT BACKLOG

Recommended implementation order:

1. Repository setup and environment configuration.
2. Authentication and organization management.
3. Tender CRUD operations.
4. Secure document upload.
5. Processing job model and status tracking.
6. PDF and spreadsheet extraction.
7. BoQ schema and validation.
8. AI extraction integration.
9. BoQ review interface.
10. Deterministic estimate calculations.
11. Supplier quotation imports.
12. Tender requirement extraction.
13. Checklist and task management.
14. Excel exports.
15. Audit logging and security tests.
16. Production deployment.
17. Pilot onboarding and measurement.

APPENDIX B — DEFINITION OF DONE

A feature is complete when:

- Its business requirements are documented.
- Input validation is implemented.
- Permissions are enforced.
- Relevant unit and integration tests pass.
- Errors are handled gracefully.
- Sensitive information is not unnecessarily logged.

- User-facing states and error messages are implemented.
- Documentation is updated.
- The feature has been tested in staging.
- Known limitations are recorded.

APPENDIX C — REFERENCES

1. South African Government, Protection of Personal Information Act 4 of 2013: https://www.gov.za/documents/protection-personal-information-act
2. South African eTenders Portal: https://www.etenders.gov.za/Home/
3. Construction Industry Development Board, Procurement Prescripts: https://www.cidb.org.za/clients/procurement-prescripts/cidb-prescripts/
4. Construction Industry Development Board, Evaluating Tenders: https://www.cidb.org.za/clients/procurement-guides/evaluating-tenders/
5. Construction Industry Development Board, Standards: https://www.cidb.org.za/contractors/advisory/cidb-standards/

APPENDIX D — FINAL SUBMISSION AND MARKING RUBRIC

- Final System Documentation with cover page containing the name of the group and names and students' number of member of the group (4 members per group)
- Final working system (the system folder and the link to access it online.)

No.	Assessment criterion	Key elements assessed	Marks
1	Background and objectives	Problem statement, objectives, target market, scope	/5
2	Requirements analysis	User stories, functional/non-functional requirements, acceptance criteria	/10
3	Architecture and database	System architecture, ERD, database relationships, API design	/10
4	UI/UX design	Dashboard, tender workspace, BoQ editor, usability, responsiveness	/8
5	Backend development	FastAPI endpoints, validation, authentication, business logic	/10
6	AI and document processing	PDF/OCR extraction, AI integration, source references, validation, accuracy	/15
7	BoQ and supplier comparison	Quantity and price calculations, quotation matching, Excel exports	/10
8	Security and compliance	Access control, tenant isolation, POPIA, secure storage, backups	/10
9	Testing and quality assurance	Unit/integration tests, AI accuracy evaluation, security and user testing	/8
10	Deployment and documentation	Deployment, version control, technical documentation, maintainability	/6
11	Business feasibility and demonstration	Pricing, market validation, customer feedback, working demonstration	/8
	TOTAL		**/100**